

1(a)(i)	<i>independent variable:</i> (type of) habitat / described ; <i>dependent variable:</i> density / number / AW, of stomata ;	2
1(a)(ii)	Radius / diameter (of field of view / image / circle / Fig 1.2) ; number of stomata ;	2
1(a)(iii)	<i>2 marks for density range:</i> 0.014–0.019 ; ; <i>if outside this range one mark for:</i> <i>stomatal number:</i> 61 (full) – 69 (all) ; <i>area:</i> 4069 – 4301 ;	2

Question	Answer	Marks
1(a)(iv)	any six from: 1 ref. to method of, random / systematic sampling of plants ; 2 (take leaves) from same species, that is present in both habitats ; 3 collect a minimum of 5 leaf samples (in both habitats) ; 4 same position / age, of leaf (on plant) ; 5 impressions to be taken from, lower / upper / both / same, surface(s) (of leaves) ; 6 (impressions viewed using) same, magnification / lens / size of field of view ; 7 count stomata (in field of view) ; 8 details of systematic counting / count only whole stomata / take a photograph and mark ; 9 repeat, (same) experiment (both habitats) minimum of twice, <u>and</u> calculate a mean ; 10 safety: hazard <u>and</u> risk <u>and</u> mitigation ;	6
1(b)(i)	73 (%) ;	1

Question	Answer	Marks
1(b)(ii)	<i>support – max 2 marks:</i> 1 stomatal density on upper surface is higher in plants grown in high light intensity ora ; 2 difference in (mean) stomatal density on upper surface (between the two light intensities) is greater than the difference between the lower surfaces ora ; 3 stomatal density on lower surface is similar / very close ; 4 leaf 1 stomatal density on lower surface is less than in plants grown in low light intensity ora ; 5 calculate pair of stomatal densities (for 1 to 4) ; <i>doesn't support – max 2 marks:</i> 6 <u>mean</u> stomatal density, on lower / both, surface(s) is <u>higher</u> in plants grown in high light intensity ora ; 7 for 2 out of 3 leaves / leaf 2 and 3, stomatal density is not higher only on the upper surface / is higher on lower / both, surface(s) ; 8 calculate pair of stomatal densities (for 6 / 7) ; ecf 9 no statistical analysis done ; 10 data only, for one species / <i>L. esculentum</i> ;	3